At the end of this worksheet you should be able to

- to discuss the relationships between the quantities of work, energy, displacement, velocity.
- differentiate between a conservative force and a non-conservative force.
- apply the work energy theorem to solve interesting problems that would be hard to use Newton's Laws.
- discuss the principle of conservation of energy and explain when it is useful.

$$\text{Work} = |\text{Force}| \cdot \cos\theta \cdot |\text{displacement}|$$

$$\Sigma F = m \cdot a$$

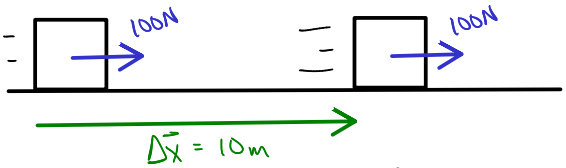
1. *Work* is defined as a transfer of *energy*. This transfer occurs by one object exerting a force on another object *over some displacement*. But the relative directions of these two vector quantities (force and displacement) matters. Summarize the work done in 5 different cases that are represented below by drawing the object and the vectors representing force $\vec{F}$ and displacement $\Delta\vec{x}$. In each case I have provided a simple example to illustrate what I mean. You provide another one.

   - The force and displacement point in exactly the same direction. (I push a box across a level floor with 100 N a distance of 10 m).

$$\text{Work} = |100\text{N}| \cdot \underbrace{\cos 0}_{1} \cdot |10\text{m}|$$

$$W = 1000\,\text{J}$$

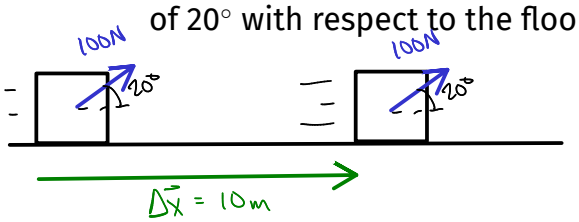
$$[\text{N}][\text{m}] \qquad \left[\frac{\text{kgm}}{\text{s}^2}\right][\text{m}] \qquad [\text{Joule}] = \left[\frac{\text{kgm}^2}{\text{s}^2}\right]$$



   - The force and displacement point in different directions, but the angle between them is less than 90°. (I pull a box across a level floor with a string, directing 100 N at an angle of 20° with respect to the floor.)

$$\text{Work} = |100\,\text{N}| \cos 20^\circ \cdot |10\,\text{m}|$$

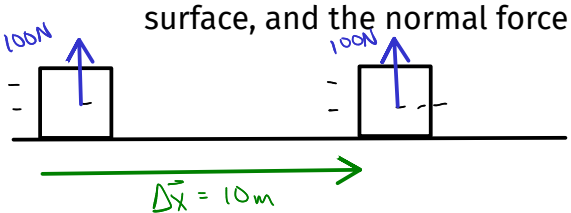
$$W = 939\,\text{J}$$



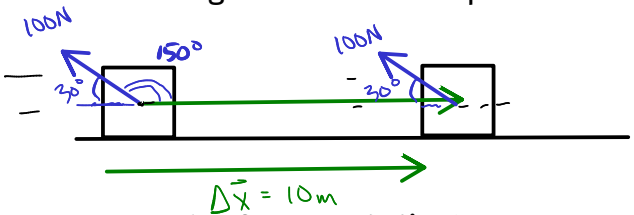
   - The force and the displacement are exactly perpendicular to each other. (A 10 kg box is sliding across a friction-less surface, and the normal force is acting on the box.)

$$W = |100\,\text{N}| \cdot \underbrace{\cos 90^\circ}_{0} \cdot |10\,\text{m}|$$

$$W = 0\,\text{J}$$



- The force and displacement point in different directions and the angle between them is greater than 90°. (I bring a sliding box to a stop by exerting a 100 N force on it at and angle of 30° with respect to the horizontal.)



$W = |100\,N| \cdot \cos(150) \cdot |10m|$

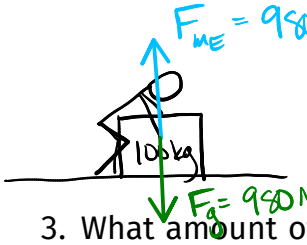
$W = -866\,J$

- The force and displacement point in exactly opposite directions. (I bring a sliding box to a stop by exerting a 100 N force over a distance of 10 m.)

$W = |100N| \cdot \underbrace{\cos(180)}_{-1} \cdot |10m|$

$W = -1000J$

2. What amount of work is done by a person to lift a 100 kg object a distance of 1 meter high? What amount of work is done by the force of gravity? If the person dropped the box what amount of work would the force of gravity do on the box as it fell? What velocity would it achieve before it hit the ground?

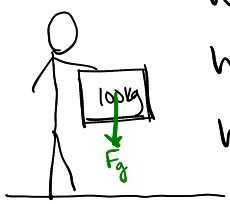


$W_{ME} = |+980N| \cos 0° |+1m|$

$W_{ME} = 980\,J$

$W_{F_g} = |-980\,N| \cos 180° |+1m|$

$W_{F_g} = -980\,J$



$W_{ME} = 0J$

$W_{F_g} = |-980N| \cos 0 \; |-1m|$

$W_{F_g} = +980J$

How fast?

$V_f^2 = V_i^{2\,\nearrow 0} + 2a\Delta y$

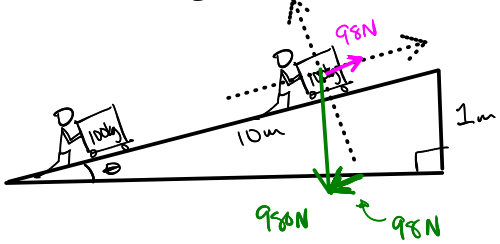
$V_f = \pm\sqrt{2a\Delta y}$

$V_f = -4.4\,m/s$

3. What amount of work is required to push a 100 kg object up a friction-less inclined plane that is 10 m long and 1 meter high at its end? How does this compare to the work done to lift it? Show that this can be used to derive the formula $\frac{F_{push}}{weight} = \frac{height}{l_{plane}}$.

$\sin\theta = \frac{1m}{10m}$

$\theta = 5.74°$



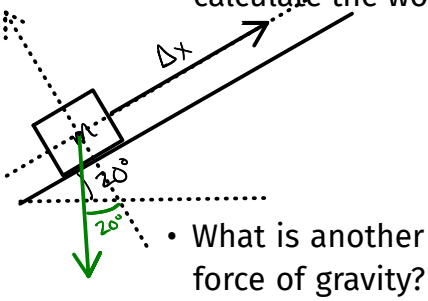
$W_{ME} = F_{ME} \cdot \cos\theta \cdot |\Delta x|$

$W_{ME} = |98N| \cdot \cos 0° \cdot |10m|$

$W_{ME} = 980J$

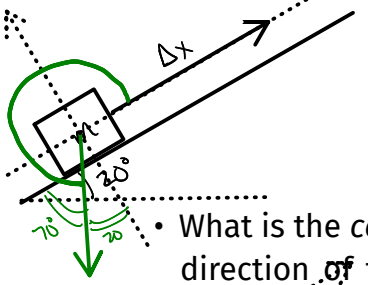
4. An object has some initial velocity at the bottom of a frictionless ramp and it begins to slide up the ramp. The force of gravity does negative work here and the object slows down to stop. The ramp has an incline angle of 20° with respect to the horizontal. Calculate the work done by the force of gravity and see that it is a negative value in three ways:

   - What is the angle between the displacement and the force of gravity? Use this angle and the definition of work to calculate the work.



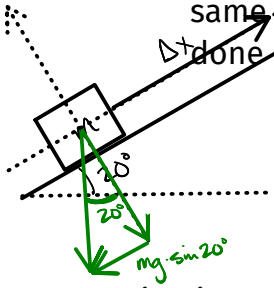
$$W_{F_g} = |F_g| \cos 110^\circ |\Delta x|$$
$$= mg \cdot \cos 110^\circ \cdot \Delta x$$
$$= -0.34 \cdot mg\Delta x$$

   - What is another angle between the displacement and the force of gravity? Now use this angle and the definition of work to calculate the work.



$$W_{F_g} = mg \cos 250^\circ \cdot \Delta x$$
$$= -0.34\, mg\Delta x$$

   - What is the *component* of the force of gravity that is in the direction of the displacement? Are these vectors in the same direction or opposite directions? What is the work done using component and the displacement?



$$W_{F_{gy}} = 0\ J$$
$$W_{F_{gx}} = |mg\underbrace{\sin 20^\circ}_{0.34}|\ \underbrace{\cos(180)}_{-1}|\Delta x|$$
$$W_{F_{gx}} = -0.34 \cdot mg\Delta x$$

5. *Kinetic energy* is the energy of an object that has velocity. In order to calculate it, use $K = \frac{1}{2}mv^2$. Calculate the kinetic energy of a 10 kg object that has a velocity of 10 m/s. If you do some work to *double* the velocity of the object, what is the new kinetic energy? What is the ratio of the kinetic energy final to the initial kinetic energy? What is the *change* in kinetic energy? How much *work* would be required to cause this change in kinetic energy? If this was done by a force pointed in the direction of the objects motion acting over a distance of 10 meters, what is the magnitude of the force?

$K = \frac{1}{2}mv^2$

$$K_i = \frac{1}{2}(10\,kg)(10\,m/s)^2$$
$$K_i = 500\ \frac{kg \cdot m^2}{s^2} = 500\,J$$

$$K_f = \frac{1}{2}(10\,kg)(20\,m/s)^2$$
$$K_f = 2000\,J$$

$$\Delta K = K_f - K_i$$
$$= 2000\,J - 500\,J$$
$$\Delta K = 1500\,J$$

$$\frac{K_f}{K_i} = \frac{2000\,J}{500\,J} = 4$$ ✓

$K \propto v^2$

$$\frac{K_f}{K_i} = \frac{v_f^2}{v_i^2} = \left(\frac{v_f}{v_i}\right)^2 = \left(\frac{20}{10}\right)^2 = 4$$ ✓

for now

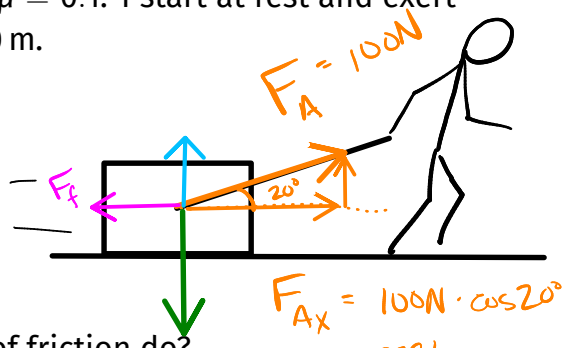
$$\Delta K = W_{total} = 1500\,J$$

$$W_{total} = W_{F_A} = |F_A| \underbrace{\cos\theta}_{1} |\Delta x|$$
$$1500\,J = F_A \cdot 10\,m$$

$F_A = 150\ N$

6. I push an object at constant velocity of 1 m/s over a friction-full surface. I exert a force of 100 N and do this over a distance of 10 m. What work have I done? What work has the force of friction done? What is the net work done? What is the kinetic energy initially? Does the kinetic energy change? What power am I providing?

7. I pull a 10 kg object with a rope at an angle of 20° to a horizontal friction-full surface. I use a force of 100 N on the rope, and the coefficient of kinetic friction is $\mu = 0.1$. I start at rest and exert this force over a distance of 100 m.



$F_{Ax} = 100N \cdot \cos 20°$
$= 93.96$

$F_{Ay} = 100N \cdot \sin 20°$
$= 34.2$

	x	y
F_g	0	$-98\ N$
F_N	0	F_N
F_f	$F_f = -6.38$	0
F_A	$+93.96$	$+34.2$
ΣF	$87.6\ N$	0

$F_N = 63.8N$

$F_f = \mu F_N$

$F_f = 6.38N$

- What work do I do?

$W_{ME} = |F_{ME}| \cdot \cos\theta \cdot |\Delta x|$

$= 100N \cdot \cos 20° \cdot 100m$

$= 9396\ J$

- What work does the force of friction do?

$W_{F_f} = |F_f| \cdot \cos\theta \cdot |\Delta x|$

$W_{F_f} = 6.38N \cdot \underbrace{\cos(180)}_{-1} \cdot 100m$

$W_{F_f} = -638\ J$

- What work does the normal force do?

$W_{F_N} = |63.8N| \cos 90° |100m|$

$W_{F_N} = 0$

- What work does the force of gravity do?

$W_{F_g} = 0$

- Using these works, what is the net work?

$$\Sigma W = 9396\,J - 638\,J + 0\,J + 0\,J = 8760\,J$$

- What is the change in kinetic energy?

$$\Sigma W = \Delta K$$
$$8760\,J = \Delta K = K_f - \cancel{K_i}^{0}$$

- What is the final velocity?

$$8760\,J = K_f = \frac{1}{2} m v_f^2$$
$$8760\,J = \frac{1}{2}(10\,kg) v_f^2$$
$$v_f = 41.8\,m/s$$

The old way w/ Newton's Laws & Kinematics

$$\Sigma F_x = ma_x$$
$$=$$
$$v_f^2 = v_i^2 + 2a\Delta x$$
$$v_f = 41.8\,m/s$$

- What is the net force?

$$U = -W_{cons}$$
$$U_g = mgh$$
$$U_s = \frac{1}{2} k \Delta x^2$$
$$U_e = \text{come back next semester}$$

- Using the net force, what is the net work?

8. What forces are conservative forces and what are not conservative forces?

Sources of potential energy:
- gravity
- spring
- electric

- friction
- applied force

9. When a 10 kg object is 10 m high, what is its potential energy? If it begins it to fall, what is its potential energy after it falls 1 m? How much has its kinetic energy changed?

$$U_{gi} = mgh_i = (10kg)(9.8\,N/kg)(10m)$$
$$U_{gi} = 980\,J$$

$$U_{gf} = m \cdot g \cdot h_f$$
$$U_{gf} = 10kg(9.8\,N/kg)(9m)$$
$$U_{gf} = 882\,J$$

$$\Delta U = U_f - U_i = 882 - 980$$
$$\Delta U = -98$$

$\Delta K + \Delta U = 0 = W_{nc}$

$$\Delta U = -\Delta K$$
$$\Delta K = +98\,J$$

Conservation of Energy

$\Delta K + \Delta U = W_{NC}$

$K_f - K_i + U_f - U_i = W_{NC}$

$K_i + U_i + W_{NC} = K_f + U_f$

10. An object falls from a height of 100 m then how fast is it going when it hits the ground? Solve this using kinematics an then again using conservation of energy?

$K_i^{\nearrow 0} + U_i + W_{NC}^{\nearrow 0} = K_f + U_f^{\nearrow 0}$

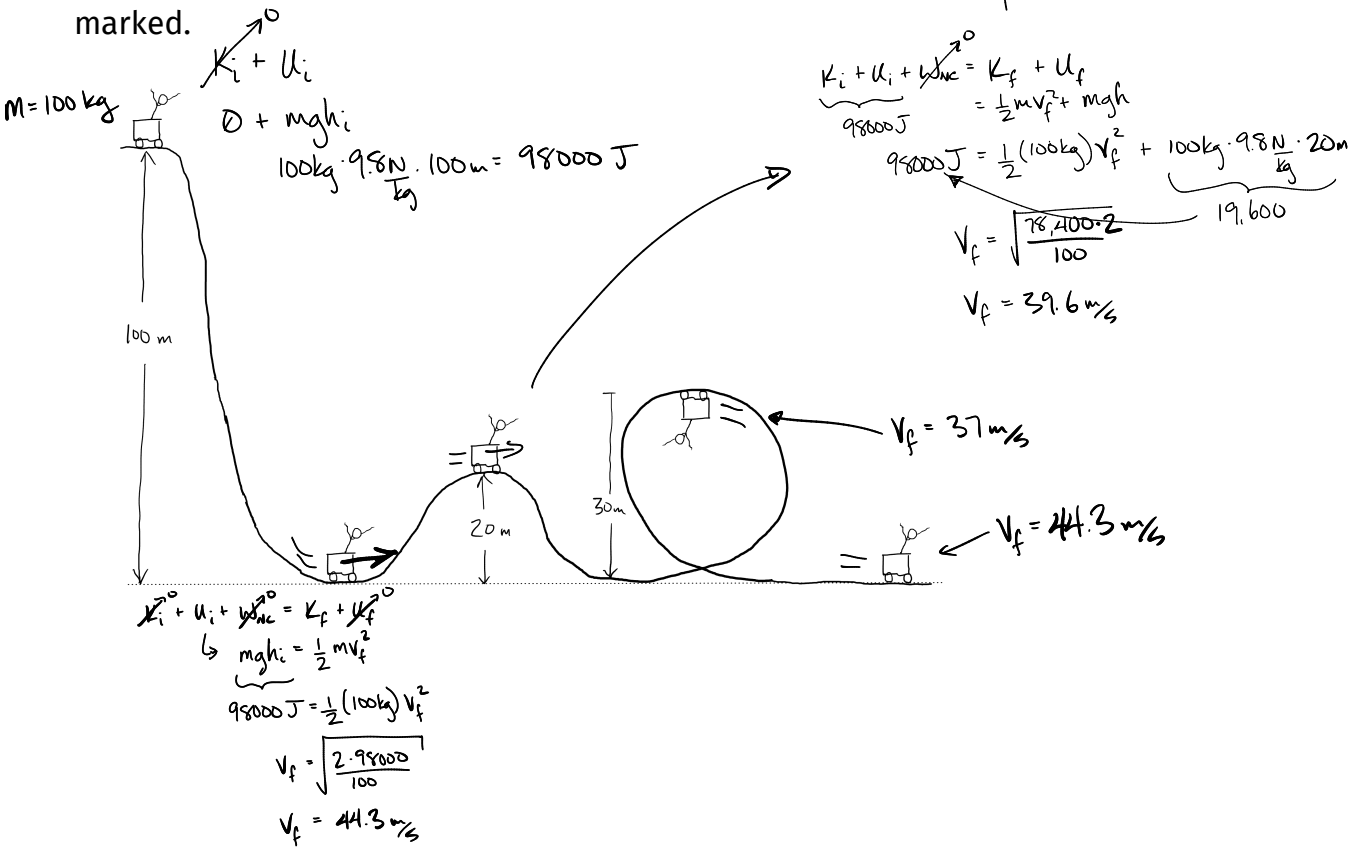
$U_i = mgh_i$ $\quad K_f = \frac{1}{2}mv_f^2$

$mgh_i = \frac{1}{2}mv_f^2$

$2gh_i = v_f^2$

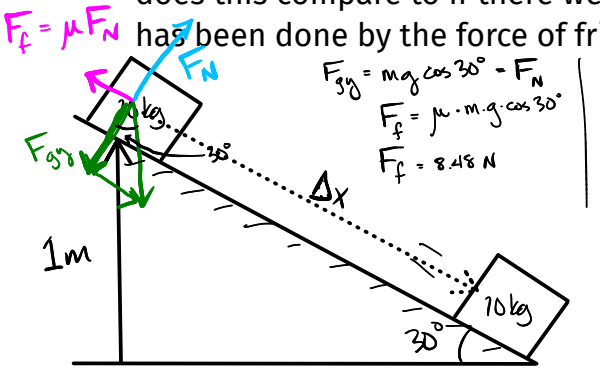
$v_f = \sqrt{2gh_i} = 44.3\ m/s$

11. A roller coaster starts from rest at the top of a hill and rolls down its course. Find its kinetic and potential energy at each position marked.



12. A 10 kg box slides down a friction-full inclined plane ($\mu = 0.1, \theta = 30°$). The height of the plane is 1 m above the horizontal. What is the speed of the box at the bottom of the plane? How

does this compare to if there were no friction? How much work has been done by the force of friction?



$F_{gy} = mg\cos 30° - F_N$

$F_f = \mu \cdot m \cdot g \cdot \cos 30°$

$F_f = 8.48\,N$

$\sin 30° = \frac{1m}{\Delta x}$

$\Delta x = \frac{1m}{\sin 30°}$

$\Delta x = 2m$

$K_i^{0} + U_i + W_{NC} = K_f + U_f^{0}$

mgh_i

$= 10 \cdot 9.8 \cdot 1m$

$= 98J$

$-17J$

$\frac{1}{2}mv_f^2$

$98J + (-17J) = \frac{1}{2}(10kg)v_f^2$

$v_f = 4.02\,m/s$

No friction, final velocity? $v_f = 4.43\,m/s$

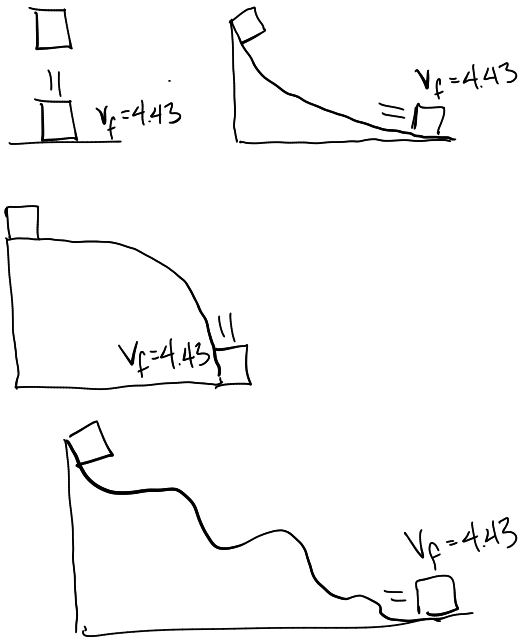
No ramp. I just drop the box.

$v_f = 4.43\,m/s$

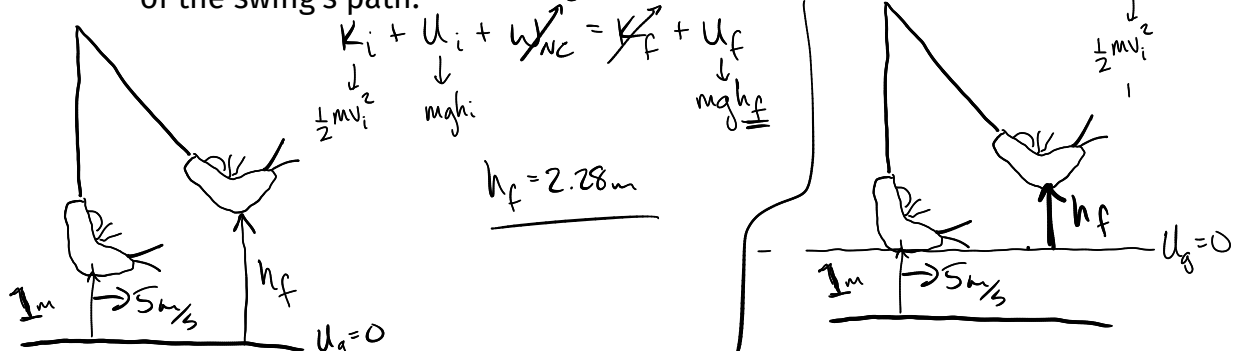
8.48N ↓ 180° ↓ 2m ↙

$W_{NC} = W_{F_f} = F_f \cdot \cos\theta \cdot \Delta x = 8.48N \cdot \cos 180° \cdot 2m = -17J$

13. Here is an example of a problem that would be much more difficult to do with Newton's Laws. Take the same inclined plane as the problem above, but make *most* of the plane friction-less and only a 10 cm portion in the middle of the plane have friction $\mu = 0.1$. Now what is the speed at the bottom of the plane? Think about how you would have solved this using Newton's Laws and kinematics.



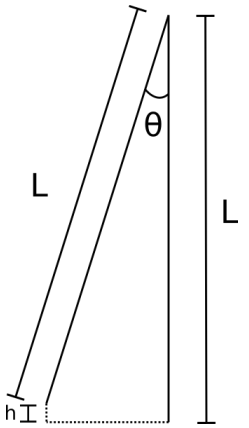
14. The maximum speed of a child on a swing is 5 m/s. At this point the child is 1 m above the ground. What is the maximum height of the child above the ground? Do this two ways, once with $U_g = 0$ at the ground, and once again with $U_g = 0$ at the bottom of the swing's path.



15. A 100 kg cyclist (and bike) has at constant speed of 10 m/s up an incline of 5% (vertical height/horizontal distance as a percent). What power output does the rider need to provide to do this?

16. A pendulum swings from some maximum height where its velocity is zero to a minimum where is velocity is a maximum and then back up to a maximum height. If the maximum height is 0.1 meters above the minimum height, then what is the speed of the pendulum at the bottom of its swing?

17. For a pendulum, it is hard to measure its maximum height, but it is easy to measure its length and to measure its angle from the vertical. If the maximum angle from the vertical of a 1 m long pendulum is 20° then how high is this above the horizontal? (*Hint: draw a line from the end of the pendulum when it is at its maximum height perpendicularly to the line when it is at is minimum height.*)



18. Three identical balls are tossed from the top of a building at the same speed. Ball A is tossed horizontally, Ball B is tossed upward at 45° , and Ball C is tossed downward.

    (a) What is each ball's total mechanical energy at the top of the building immediately after the toss?

    (b) What is the speed of each ball as it strikes the ground?

    (c) Will energy tell us when the ball lands?

    (d) Will energy tell us how far each ball lands from the base of the building?

19. A soccer ball is kicked across a field with a velocity of 5 m/s at 37° above the horizontal. How can you determine the maximum height of the ball using energy?

20. What is the minimum height h that a roller coaster needs to start at rest in order to do a loop-the-loop of radius r and not lose contact with the track?

21. A 2 kg mass rests at the top of a frictionless curved ramp 3 m above its base. The mass slides down a frictionless curved ramp and strikes a spring with a spring constant of 20 N m^{-1}. What distance is the spring compressed when the block comes to rest?

